

SQL PROJECT REPORT

Analysis of Netflix Global Content Library

Using PostgreSQL 18 & pgAdmin 4

Submitted by: Tanisha

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Dataset Source: Kaggle's Netflix Movies and TV Shows

Tool: PostgreSQL 18 / pgAdmin 4

1. Introduction

This project presents a structured SQL-based analysis of the Netflix Movies and TV Shows dataset, sourced from Kaggle. The dataset comprises 8,807 records of Netflix content, spanning movies and television shows from across the globe and includes attributes such as content type, country of production, genre, director, content rating, duration, and date of addition to the platform.

The project was executed entirely in PostgreSQL 18 using pgAdmin 4 as the query interface. It demonstrates the practical application of SQL from foundational data retrieval and cleaning through to advanced analytical functions in the context of a real-world entertainment dataset. The analysis mirrors the kind of data-driven decision-making that streaming platforms like Netflix rely on to manage content acquisition, audience targeting, and regional expansion strategies.

2. Aim

To explore, clean, and analyze Netflix's global content library using PostgreSQL and examining patterns in content type, geography, genre, ratings, and growth over time, in order to derive meaningful insights into how Netflix has evolved as a global streaming platform and how data can inform content strategy decisions.

3. Objectives

- Design a structured and normalized table schema in PostgreSQL to store Netflix content data efficiently
- Import and load the raw CSV dataset into pgAdmin using the platform's built-in data import utility
- Perform comprehensive data cleaning, i.e., identifying and handling NULL values, duplicate entries, whitespace inconsistencies, and data entry errors
- Profile the dataset by analyzing distributions across content type, country of origin, and content rating
- Study content growth trends on the platform year over year to understand Netflix's content investment timeline
- Identify dominant genres and examine their relationship with content type across the library
- Analyze top directors and quantify their contribution to the Netflix platform by volume and versatility
- Examine movie duration patterns and TV show season lengths to understand content format preferences

- Investigate audience targeting strategies through the analysis of content rating distributions
- Draw actionable business insights reflecting Netflix's content acquisition strategy, regional partnerships, and global expansion

4. Dataset Overview

The dataset used in this project is the Netflix Movies and TV Shows dataset published on Kaggle. It is a publicly available, single-file CSV dataset containing comprehensive metadata for every title available on Netflix at the time of collection. The table below summarizes the dataset's key characteristics:

Attribute	Details
Dataset Name	Netflix Movies and TV Shows
Source	Kaggle (kaggle.com/datasets/shivamb/netflix-shows)
Total Records	8,807 rows
Total Columns	12
File Format	CSV (Comma Separated Values)
Database	PostgreSQL 18
Query Interface	pgAdmin 4

4.1 Table Schema

The following schema was designed to accommodate all 12 attributes of the dataset. Data types were selected carefully. BIGINT for large numeric fields, VARCHAR with appropriate lengths for text, and TEXT for open-ended descriptions:

Column Name	Data Type	Description
show_id	VARCHAR(10)	Unique identifier for each title
type	VARCHAR(10)	Content type: Movie or TV Show
title	VARCHAR(200)	Title of the content
director	VARCHAR(300)	Director name — nullable field
casts	VARCHAR(1000)	Comma-separated cast member names
country	VARCHAR(200)	Country of production — nullable
date_added	VARCHAR(50)	Date the title was added to Netflix
release_year	INT	Year of original release
rating	VARCHAR(10)	Content rating: TV-MA, PG, R, etc.

Column Name	Data Type	Description
duration	VARCHAR(20)	Duration in minutes or number of seasons
listed_in	VARCHAR(300)	Genre tags associated with the title
description	TEXT	Plot summary or description

5. Data Cleaning and Pre-Processing

Before any analytical queries were executed, the dataset underwent a structured cleaning process. This phase is critical in any data project. Raw data from real-world sources invariably contains inconsistencies, missing values, and formatting issues that can distort analysis results if left unaddressed.

5.1 NULL Value Analysis

The first cleaning step was to quantify missing data across all key columns. The result revealed the following:

Column	NULL Count	% of Total	Implication
director	2,634	~30%	Common for TV shows — no single director credited
country	831	~9.4%	Some titles lack geographic origin data
rating	4	~0.05%	Negligible — minor data entry gaps
date_added	10	~0.11%	Very minor — a few titles have no addition date

The 30% NULL rate in the director column is not a data quality failure but is a structural characteristic of the dataset. TV shows rarely attribute a single director, and thus this field is legitimately absent for a large portion of entries. This observation itself constitutes a meaningful insight about how content metadata is maintained.

5.2 Duplicate Detection

A query was run to identify titles appearing more than once in the dataset. Approximately 200 titles were found with duplicate names. Upon closer inspection, these were not true duplicates. They represented the same title available in multiple regions, or different versions of the same content. No records were deleted as a result, confirming the dataset's integrity at the record level.

5.3 Data Entry Error : Rating Column

A significant data quality issue was uncovered during the unique ratings analysis: three entries which are '66 min', '74 min', and '84 min' appeared in the rating column. These are clearly duration values that were incorrectly entered into the wrong field. This finding demonstrates the importance of exploratory data analysis prior to drawing conclusions, as such errors can silently corrupt aggregate results if undetected.

5.4 Whitespace Trimming

An UPDATE query was applied across all primary text columns like title, director, country, and rating to remove leading and trailing whitespace. This is a standard but essential step, as whitespace inconsistencies cause incorrect GROUP BY results, failed WHERE matches, and JOIN failures in multi-table environments. The operation successfully updated all 8,807 rows.

6. Analytical Framework and Query Design

The 28 SQL queries in this project were not written arbitrarily, rather were organized into four deliberate tiers, each building on the previous, to reflect a logical analytical progression from data exploration to business intelligence. This structure mirrors how professional data analysts approach a new dataset.

6.1 Tier 1 — Basic Queries (Queries 1–7): Data Exploration

The first tier focused on understanding the shape and composition of the dataset. Queries at this level used SELECT, DISTINCT, WHERE, ILIKE, GROUP BY, ORDER BY, and LIMIT the foundational building blocks of SQL. The intent was to answer the most immediate questions: What does this data look like? How is it distributed? Are there obvious patterns or problems?

For example, the very first analytical query counted Movies versus TV Shows using GROUP BY on the type column. This single query immediately established the most important structural fact about the dataset is that Netflix's library is 70% movies, a finding that frames all subsequent analysis. Similarly, a pattern-matching query using the ILIKE operator with the wildcard %love% demonstrated how SQL powers real-world content search features, mirroring the search functionality users experience on the Netflix interface itself.

6.2 Tier 2 — Intermediate Queries (Queries 8–17): Aggregation and Pattern Finding

The second tier moved from description to analysis using aggregate functions (AVG, SUM, COUNT, MAX, MIN), date extraction functions (EXTRACT, TO_CHAR, TO_DATE), string manipulation functions (SPLIT_PART, CAST), and conditional filtering (HAVING, multiple AND/OR conditions) to uncover patterns within the data.

This tier addressed more sophisticated questions: How has Netflix's content library grown over time? Which month sees peak content additions? What is the average duration of a Netflix movie? Which directors have contributed the most titles? These queries required combining multiple SQL concepts in a single statement, reflecting the complexity of real analytical tasks. The year-over-year content trend query, for instance, required extracting year values from a VARCHAR date column, grouping by those extracted values, and ordering chronologically a multi-step operation that tests practical SQL competence.

6.3 Tier 3 — Advanced Queries (Queries 18–25): Deep Analysis

The third tier introduced the most powerful SQL constructs used in the project. Window functions, subqueries, UNION operations, CASE WHEN conditional logic, and INTERVAL-based date arithmetic. These are the tools that distinguish basic SQL users from analytical practitioners.

The year-over-year growth query used the LAG() window function to calculate the difference in content additions between consecutive years a technique directly borrowed from time-series analysis. The director ranking query employed RANK() OVER (PARTITION BY country) to rank directors within each country independently, without collapsing the dataset through aggregation. The India vs USA comparison query used CASE WHEN to segment data dynamically, and the FILTER clause to compute conditional aggregates producing a side-by-side analytical comparison in a single query execution.

6.4 Tier 4 — Bonus Queries (Queries 26–28): Business Intelligence

The final tier translated raw analytical findings into business-relevant questions. These queries were designed to surface insights that a content strategy team or marketing analyst at Netflix would genuinely care about which small countries punch above their weight in mature content production, where hidden opportunities exist in the mid-tier of the influencer ranking, and how the three biggest English-speaking markets compare head to head. This tier demonstrated that SQL is not merely a data retrieval tool but can also work as a business intelligence engine when wielded thoughtfully.

7. Key Findings and Interpretation

The following section presents the most significant findings derived from the 28-query analysis, organized thematically rather than by query number. Each finding is supported by the data and contextualized within Netflix's known business strategy.

7.1 Content Composition — Movies Dominate

Netflix's library consists of 6,131 movies (69.62%) and 2,676 TV shows (30.38%). This 70/30 split is one of the most counterintuitive findings of the project popular culture associates Netflix strongly with prestige TV productions, yet the platform's actual library is overwhelmingly movie-heavy. This reflects Netflix's historical strategy of licensing large volumes of existing film catalogues to build initial subscriber bases, while TV show originals generate cultural conversation but represent a smaller share of total titles.

7.2 Geographic Distribution — The USA Leads, India Punches Above Its Weight

The USA accounts for 2,818 titles which is nearly three times India's 972 and almost seven times the UK's 419. However, India's strong second-place position is particularly significant given the relatively recent globalization of its film industry on international platforms. The presence of Egypt, Mexico, South Korea, and Spain in the top 10 reflects Netflix's deliberate investment in regional content markets, driven by the understanding that local-language content drives subscriber growth in non-English markets.

7.3 Content Growth Timeline — The 2017 Inflection Point

Netflix added fewer than 100 titles per year between 2008 and 2015. This changed dramatically in 2016, with 429 additions, followed by a jump of 759 titles in 2017 alone — the largest single-year growth recorded in the dataset. The peak was reached in 2019 with 2,016 titles added in a single year. This growth curve maps precisely onto Netflix's publicly reported content budget expansions, which grew from approximately \$5 billion in 2016 to over \$15 billion by 2019. The data tells the story of a company that deliberately transformed itself from a content distributor into a content producer.

7.4 Audience Targeting — Adult Content Dominates

TV-MA (Mature Audiences, 17+) is the single most common content rating for both movies (2,062 titles) and TV shows (1,145 titles). This finding confirms that Netflix's primary target demographic is adults. The platform's most acclaimed and discussed content ranging from Breaking Bad to Squid Game to Narcos, falls within this rating category, and the data validates that this is not coincidental but strategic.

7.5 Content Format Patterns

The average Netflix movie runs exactly 100 minutes. Perfectly aligned with the global theatrical release standard. This consistency suggests that Netflix's movie licensing and production decisions are calibrated to match viewer expectations shaped by cinema. On the TV side, 259 shows have more than three seasons, with Grey's Anatomy leading at 17 seasons indicating that Netflix's catalogue includes a significant library of established long-running franchises that provide reliable, low-risk viewing volume.

7.6 Seasonal Strategy — July is Peak Month

July recorded the highest number of content additions in the dataset, with 827 titles. This is strategically deliberate as we know summer months correspond to peak leisure time and holidays for Netflix's primary Western markets, and front-loading content releases in July maximizes the audience available to consume new additions. This finding demonstrates how even temporal metadata in a dataset can reveal platform-level business strategy.

7.7 Same-Year Acquisition — Fresh Content Investment

Approximately 1,000 titles representing, 11% of the library were added to Netflix in the same calendar year they were originally released. This indicates that Netflix actively acquires a meaningful volume of content directly from production, rather than relying solely on catalogue licensing. This fresh-content pipeline is central to Netflix's value proposition as a platform that offers new releases, not just archive viewing.

8. Interpretation Matrix

The table below provides a consolidated summary of all 28 queries along with their analytical purpose, the SQL concepts demonstrated, and the key insight derived from each result.

Interpretation Matrix



Query No.	Query Table	SQL Concept Used	Key Result	Observation
1	View full dataset	SELECT *	8807 rows, 12 columns	Dataset contains 8807 Netflix titles across 12 attributes including type, country, rating, and duration
2	Movies vs TV Shows count	GROUP BY, COUNT	Movies: 6131, TV Shows: 2676	Netflix has 2.3x more movies than TV shows 69.62% of content is movies, challenging the common perception of Netflix as a TV show platform
3	Unique content ratings	SELECT DISTINCT	18 distinct values	3 dirty entries (66 min, 74 min, 84 min) found in the rating column a data quality issue where duration values were incorrectly entered as ratings
4	Missing director info	WHERE IS NULL	2634 NULL values	Nearly 30% of all titles have no director information. This is common for TV shows where no single director is credited
5	Indian content on Netflix	WHERE ILIKE	196 titles	India contributes 196 titles making it the 2nd largest content producing country on Netflix, reflecting Bollywood's global reach
6	Keyword search — "Love" titles	ILIKE with % wildcard	Multiple titles found	The %love% wildcard matches the word in any position in the title — this mirrors how real-world streaming search engines work using pattern matching
7	Top 10 content producing countries	GROUP BY, ORDER BY, LIMIT	USA: 2818, India: 972, UK: 419	The USA produces 3x more content than India and 7x more than the UK — however India's strong 2nd place shows Netflix's significant investment in regional content
8	Content added per year	EXTRACT, GROUP BY	Peak in 2019 with 2016 titles	Netflix's content library grew slowly until 2016, then exploded — 2017 saw a jump of 759 titles reflecting Netflix's billion-dollar content spending strategy
9	Most frequent rating per type	GROUP BY type + rating	TV-MA dominates both	TV-MA is the #1 rating for both Movies and TV Shows — Netflix primarily targets adult audiences aged 17 and above

10	Top 10 most prolific directors	GROUP BY director, HAVING	Rajiv Chilaka tops the list	Rajiv Chilaka (creator of Chhota Bheem) is Netflix's most prolific director — driven by the large number of animated specials in the franchise
11	Average movie duration	AVG, SPLIT_PART, CAST	100 minutes	The average Netflix movie is exactly 100 minutes long — this aligns with the industry standard of 90–120 minutes for theatrical releases
12	TV Shows with 3+ seasons	SPLIT_PART, CAST, WHERE	259 shows, Grey's Anatomy tops at 17 seasons	259 TV shows on Netflix have more than 3 seasons — Grey's Anatomy leads with 17 seasons, showing Netflix hosts well-established long-running franchises
13	Most common genres	GROUP BY <u>listed_in</u>	Dramas + International Movies: 362	Drama is Netflix's dominant genre category — the high <u>count</u> of International Movies confirms Netflix's strong push for global and regional content
14	Content additions by month	TO_CHAR, EXTRACT	July tops with 827 titles	July is Netflix's biggest content addition month — likely timed with summer holidays when viewership is highest
15	TV-MA content after 2018	WHERE with multiple conditions	Large result set	A significant volume of mature content was added post-2018 — reflecting Netflix's growing focus on adult-oriented original programming
16	Countries with both Movies and TV Shows	COUNT DISTINCT, HAVING	Multiple countries	Several countries produce both content types for Netflix — showing the platform's diverse international production partnerships
17	Percentage split Movies vs TV Shows	Subquery, arithmetic	Movies: 69.62%, TV: 30.38%	Despite TV shows generating more cultural buzz, they represent only 30% of Netflix's total library — movies form the backbone of the platform's content strategy
18	Duplicate titles	GROUP BY, HAVING COUNT > 1	200 similar titles	200 titles appear more than once — these are not true duplicates but represent the same title available in different regions or versions
19	Same-year acquisition	EXTRACT, WHERE	~1000 titles	Approximately 11% of Netflix content was added in the same year it was produced — indicating Netflix actively acquires fresh content straight from production

20	Year-over-year growth	LAG, subquery	2017 had biggest jump: +759	2017 marks Netflix's most aggressive content expansion year — this directly correlates with Netflix's reported \$6 billion content budget announced that year
21	Top genre by content type	GROUP BY type + <u>listed_in</u>	Dramas lead for both types	International Movies and Dramas <u>dominate</u> for Movies; TV Dramas lead for TV Shows — drama is Netflix's universal content language across all formats
22	Directors with Movies and TV Shows both	COUNT DISTINCT type, HAVING	Multiple versatile directors	Several directors have worked across both formats on Netflix — showing the platform encourages creative flexibility among its content partners
23	Longest and shortest movies	Subquery with MAX and MIN	Longest: 312 min, Shortest: 3 min	Netflix hosts extremely diverse movie lengths — from 3-minute short films to 5+ hour documentaries, catering to all viewing preferences
24	Content added in last 3 years	WHERE with INTERVAL	Large recent dataset	A significant portion of Netflix's library is recent content — confirming the platform continuously refreshes its catalogue to retain subscribers
25	India vs USA comparison	CASE WHEN, GROUP BY	USA leads in volume, India strong in movies	USA dominates in total titles but India shows strong movie production numbers — both countries have similar average movie durations around 100 minutes
26	Top directors per country	<u>RANK()</u> OVER PARTITION BY	Country-wise rankings revealed	Using window functions, top directors were ranked within each country — Rajiv Chilaka leads India, showing how regional creators dominate their local Netflix catalogues
27	Hidden gems — small countries with mature content	HAVING COUNT BETWEEN	Multiple niche countries	Several smaller countries with 10–50 titles on Netflix produce a disproportionately high share of mature content suggesting niche adult-oriented regional productions
28	India vs USA vs UK three-way battle	CASE WHEN, GROUP BY	USA: highest titles, India: 2nd, UK: 3rd	USA leads with the highest content volume but India and UK are strong competitors, average release years across all three are similar, showing Netflix globally sources both old and new content

9. Conclusion

This project successfully demonstrated that SQL is far more than a data retrieval language. It is a full-spectrum analytical tool capable of powering real-world business intelligence when applied systematically and thoughtfully. Through 28 structured queries across four analytical tiers, the Netflix dataset was transformed from raw tabular data into a coherent narrative about one of the world's most influential media companies.

The analysis revealed several findings of genuine strategic relevance: Netflix's library is movie-dominated despite its TV show reputation; the USA leads content production but India's second-place position reflects a deliberate regional investment strategy; the year 2017 marks the decisive inflection point in Netflix's content expansion; adult audiences are the platform's primary target demographic; and July is the platform's preferred month for content releases, aligned with peak viewership periods.

From a technical standpoint, the project covered the full spectrum of SQL competency ranging from foundational SELECT and GROUP BY operations, through intermediate aggregation and date manipulation, to advanced window functions (RANK, LAG), subqueries, CASE WHEN logic, and string parsing functions (SPLIT_PART, ILIKE, TO_CHAR). The data cleaning phase further demonstrated the practical importance of pre-processing in any data analysis pipeline.

Ultimately, this project illustrates that the value of SQL lies not in the syntax itself, but in the quality of the questions asked. Every query in this analysis was motivated by a real business question and the answers, taken together, form a data-driven portrait of how Netflix has built and manages the world's largest streaming content library.

10. References

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